| | |
|---|---|
| (ii) **The commutative property**<br>For any two complex numbers $z_1$ and $z_2$<br>$z_1 + z_2 = z_2 + z_1$. | (ii) **The commutative property**<br>For any two complex numbers $z_1$ and $z_2$<br>$z_1 z_2 = z_2 z_1$. |
| (iii) **The associative property**<br>For any three complex numbers $z_1, z_2$, and $z_3$<br>$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$. | (iii) **The associative property**<br>For any three complex numbers $z_1, z_2$, and $z_3$<br>$(z_1 z_2) z_3 = z_1 (z_2 z_3)$. |
| (iv) **The additive identity**<br>There exists a complex number $0 = 0 + 0i$ such that, for every complex number $z$,<br>$z + 0 = 0 + z = z$<br>The complex number $0 = 0 + 0i$ is known as additive identity. | (iv) **The multiplicative identity**<br>There exists a complex number $1 = 1+0i$ such that, for every complex number $z$,<br>$z1 = 1z = z$<br>The complex number $1 = 1 + 0i$ is known as multiplicative identity. |
| (v) **The additive inverse**<br>For every complex number $z$ there exists a complex number $-z$ such that,<br>$z + (-z) = (-z) + z = 0.$<br>$-z$ is called the additive inverse of $z$. | (v) **The multiplicative inverse**<br>For any nonzero complex number $z$, there exists a complex number $w$ such that,<br>$z w = w z = 1.$<br>$w$ is called the multiplicative inverse of $z$.<br>$w$ is denoted by $z^{-1}$. |
| (vi) **Distributive property (multiplication distributes over addition)**<br>For any three complex numbers $z_1$, $z_2$, and $z_3$<br>$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$ and $(z_1 + z_2) z_3 = z_1 z_3 + z_2 z_3$. | |

Let us now prove some of the properties.

**Property**

The commutative property under addition

For any two complex numbers $z_1$ and $z_2$, we have $z_1 + z_2 = z_2 + z_1$.

**Proof**

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, and $x_1, x_2, y_1$, and $y_2 \in \mathbb{R}$,

$$\begin{aligned} z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) \\ &= (x_1 + x_2) + i(y_1 + y_2) \\ &= (x_2 + x_1) + i(y_2 + y_1) \qquad (\text{since } x_1, x_2, y_1, \text{ and } y_2 \in \mathbb{R}) \\ &= (x_2 + iy_2) + (x_1 + iy_1) \\ &= z_2 + z_1. \end{aligned}$$

**Property** ■

Inverse Property under multiplication

Prove that the multiplicative inverse of a nonzero complex number $z = x + iy$ is $\frac{x}{x^2 + y^2} + i\frac{-y}{x^2 + y^2}$.



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